Unit: Unit 0: Foundations of Algebra
Topic: Rational Numbers Decimals

Name: ____________

Date: ________

Advanced (Carolina Reaper)

Q1. A music store sells guitars for

$$\frac{3}{4}$$

of the original price. If the original price is $800, how much will you pay?$

Q2.

$$600$$

Q3.

$$640$$

Q4.

$$540$$

Q5.

$$580$$

Q6.

$$620$$

What is

$$0.\overline{45}$$

as a fraction?
(A)

$$\frac{1}{20}$$

(B)

$$\frac{5}{11}$$

(C)

$$\frac{1}{22}$$

(D)

$$\frac{41}{90}$$

(E)

$$\frac{4}{9}$$

A bakery is having a sale on bread. A loaf that originally costs

$$\frac{3}{5}$$

of a dollar is now on sale for

$$0.6$$

dollars. What is the percent decrease in price?
(A)

$$20\%$$

(B)

$$10\%$$

(C)

$$5\%$$

(D)

$$15\%$$

(E)

$$25\%$$

Order the following rational numbers from least to greatest:

$$\frac{5}{8}, 0.6, \frac{9}{16}, 0.7$$

(A)

$$0.6, \frac{5}{8}, \frac{9}{16}, 0.7$$

(B)

$$0.6, \frac{9}{16}, \frac{5}{8}, 0.7$$

(C)

$$\frac{9}{16}, 0.6, \frac{5}{8}, 0.7$$

(D)

$$0.6, \frac{9}{16}, 0.7, \frac{5}{8}$$

(E)

$$\frac{5}{8}, \frac{9}{16}, 0.6, 0.7$$

If

$$x = 0.\overline{9}$$

, then

$$x + \frac{1}{9}$$

equals
(A)

$$\frac{10}{9}$$

(B)

$$1$$

(C)

$$\frac{11}{9}$$

(D)

$$2$$

(E)

$$\frac{12}{9}$$

A water tank can hold

$$\frac{3}{4}$$

of a thousand gallons. If

$$\frac{1}{6}$$

of the tank is filled, how many gallons of water are in the tank?
(A)

$$50$$

gallons
(B)

$$40$$

gallons
(C)

$$60$$

gallons
(D)

$$125$$

gallons
(E)

$$75$$

gallons

Open Ended Questions (FRQ)

Q9. Prove that

$$0.\overline{9} = 1$$

using a geometric series and show your work.

Q10. Graph the rational numbers

$$-\frac{5}{6}, -0.8, \frac{1}{3}, 0.5$$

on a number line and label each point. Show your scale.

Chef's Tips

- Emphasize that the format of the response is crucial for FRQ questions.
- Highlight the importance of explaining the reasoning behind the answer.
- Encourage the use of diagrams and visual aids where necessary.